Software Engineering — LAB 2

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Student Reference Number (SRN)	PES1UG24CS396
Lab Scenario Number	Problem Statement #28
Primary Domain	Smart Cities, Transport & Logistics

EPICS

1. **Freight Load Management & Bidding** — Enable shippers to list loads and carriers to discover and bid on freight shipments.
2. **Shipment Tracking & Escrow Settlement** — Provide real-time milestone tracking and automated escrow payment disbursement upon delivery verification.
3. **Platform Quality, Scalability & Compliance Verification** — Validate non-functional SLA performance (<60s escrow release) and 500+ session concurrency benchmarks.

Epics Overview

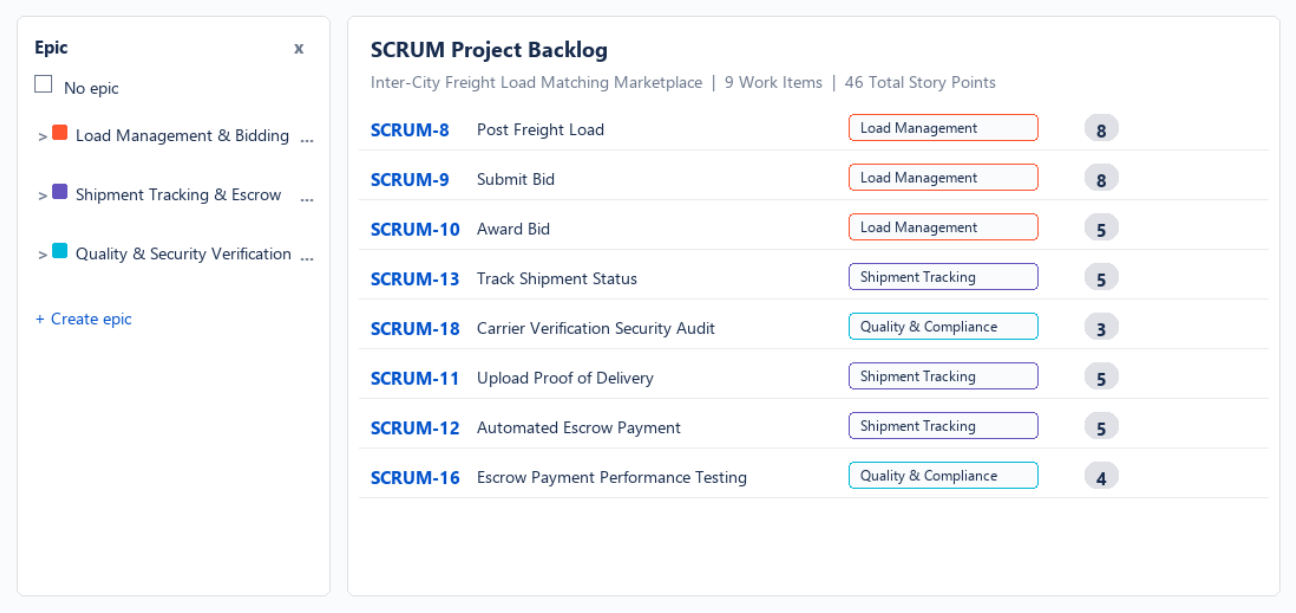


Figure 1: Epics and Backlog panel configured in Jira for Problem Statement #28.

Epic: Freight Load Management & Bidding

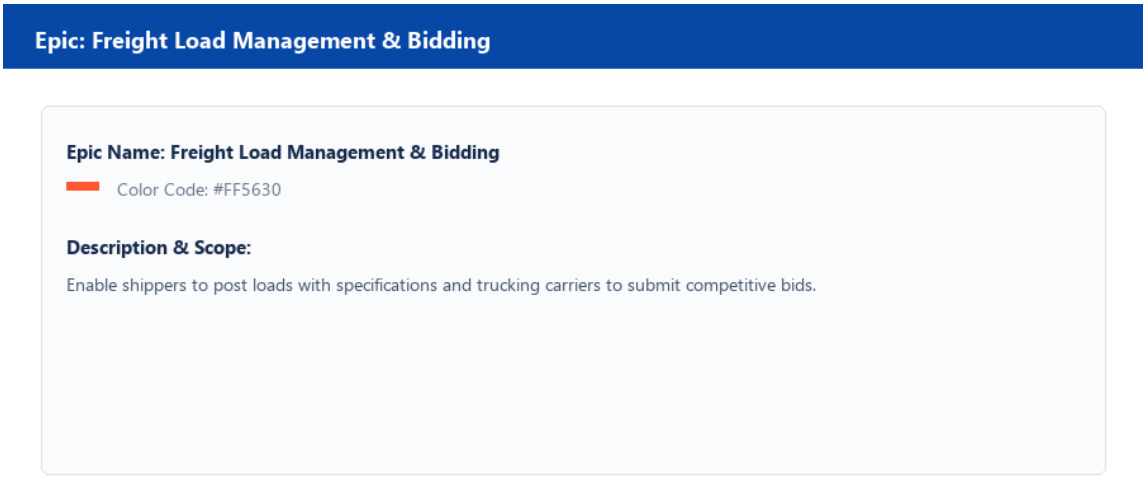


Figure 2: Scope and metadata configuration for Epic 1.

Epic: Shipment Tracking & Escrow Settlement

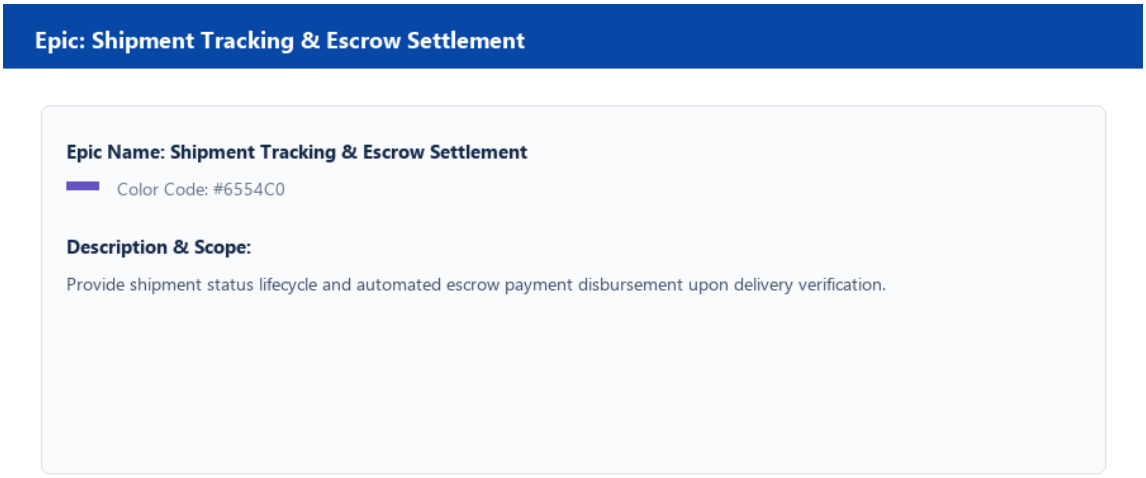


Figure 3: Scope and metadata configuration for Epic 2.

Complete Backlog

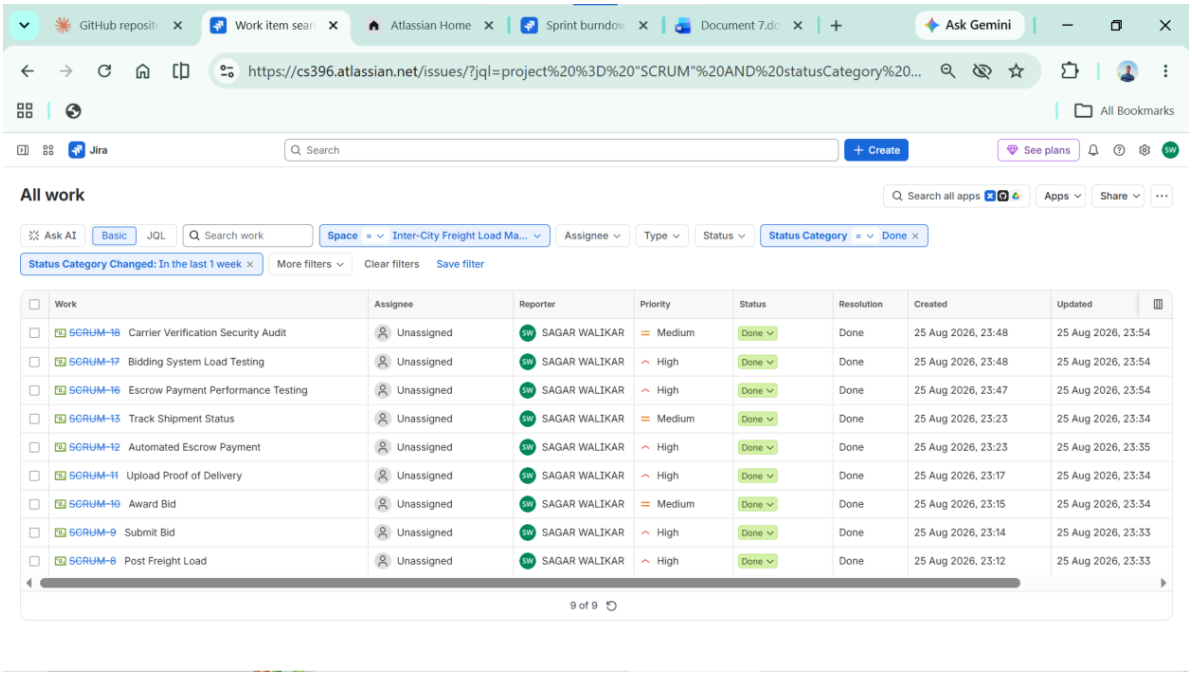


Figure 4: Complete product backlog showing all 9 work items (SCRUM-8 to SCRUM-18).

Prioritized Backlog

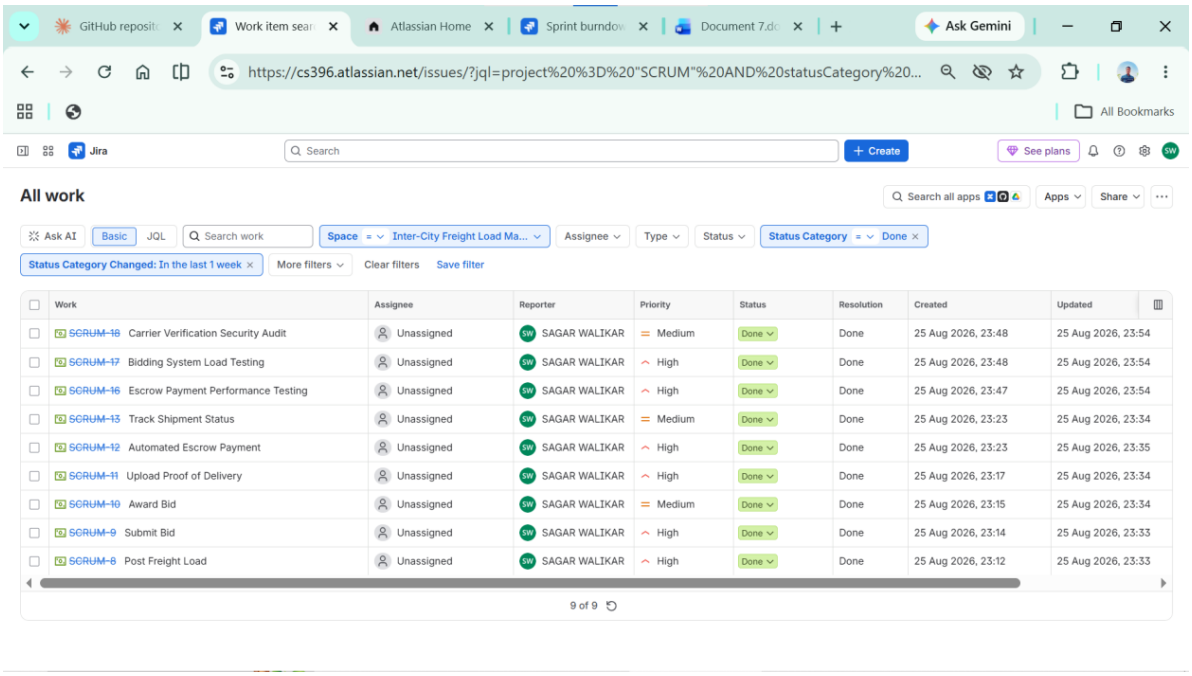


Figure 5: Backlog prioritized by technical dependencies and architectural risk.

Story Points Estimation

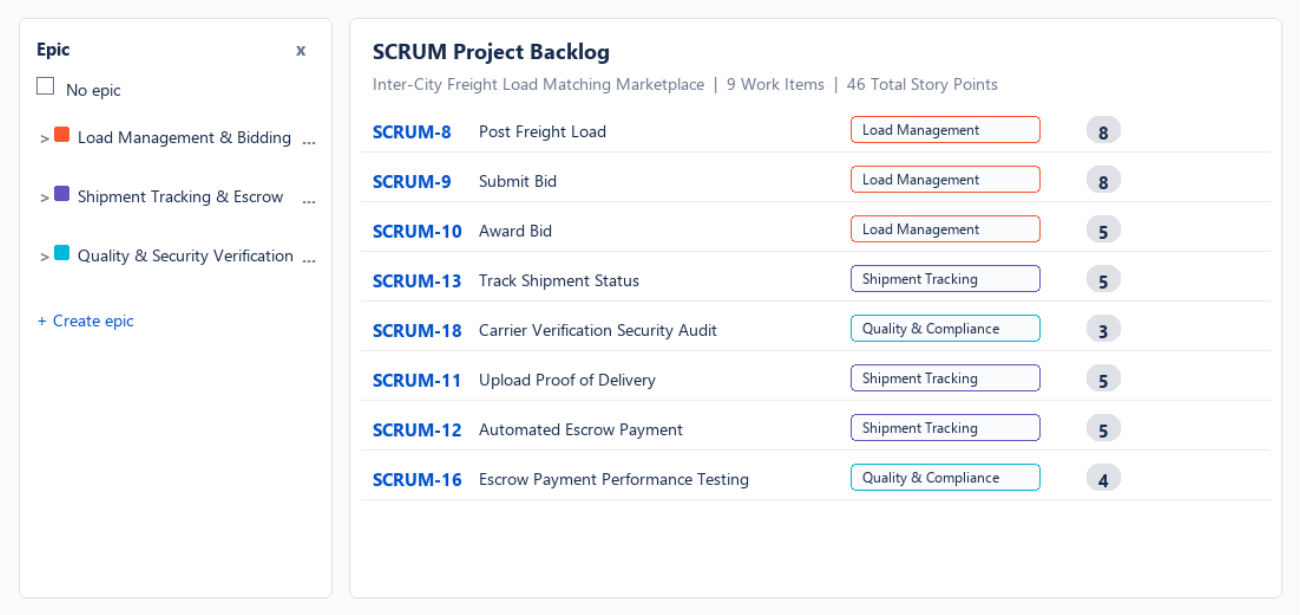


Figure 6: Fibonacci story point assignments across all backlog items (Total: 46 Story Points).

Sprint 1 Planning

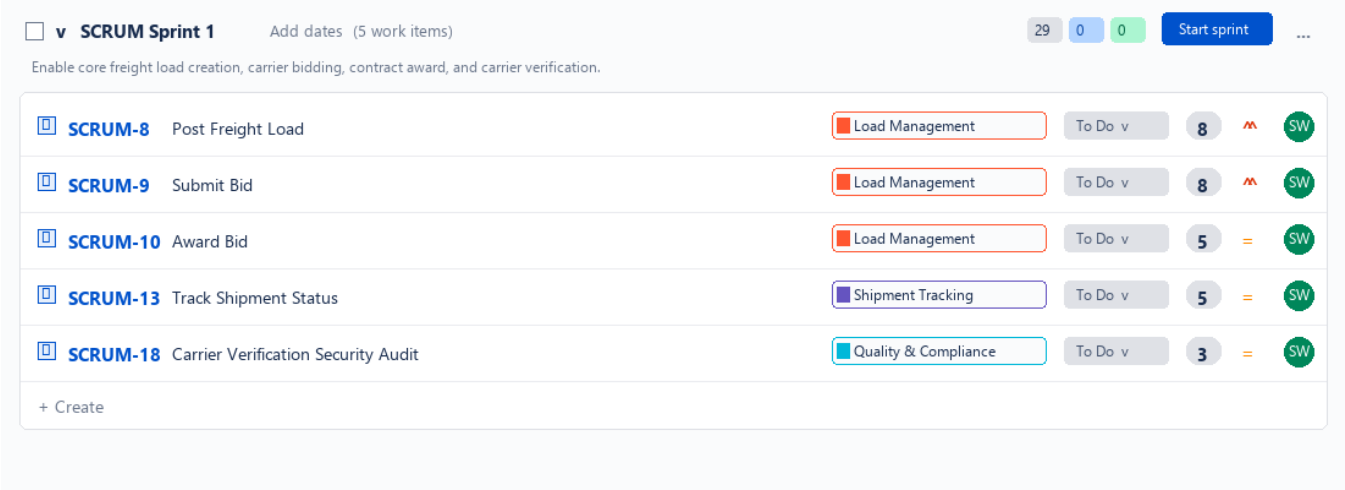


Figure 7: Sprint 1 scope: 5 work items totaling 29 story points.

Sprint 1 — To Do

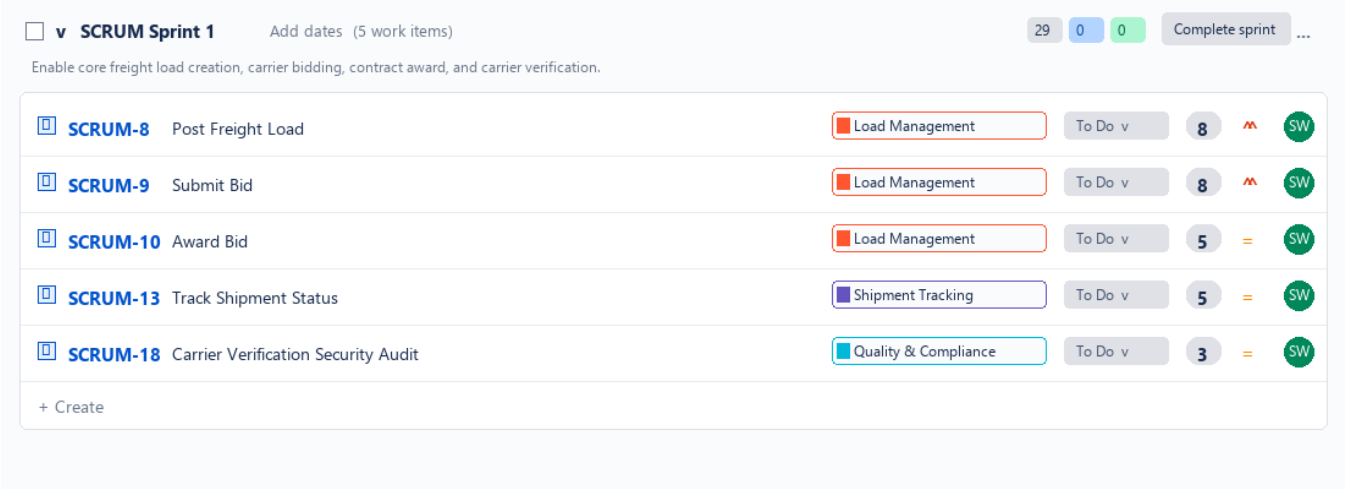


Figure 8: Sprint 1 active board at sprint initiation with work items in To Do.

Sprint 1 — Progress

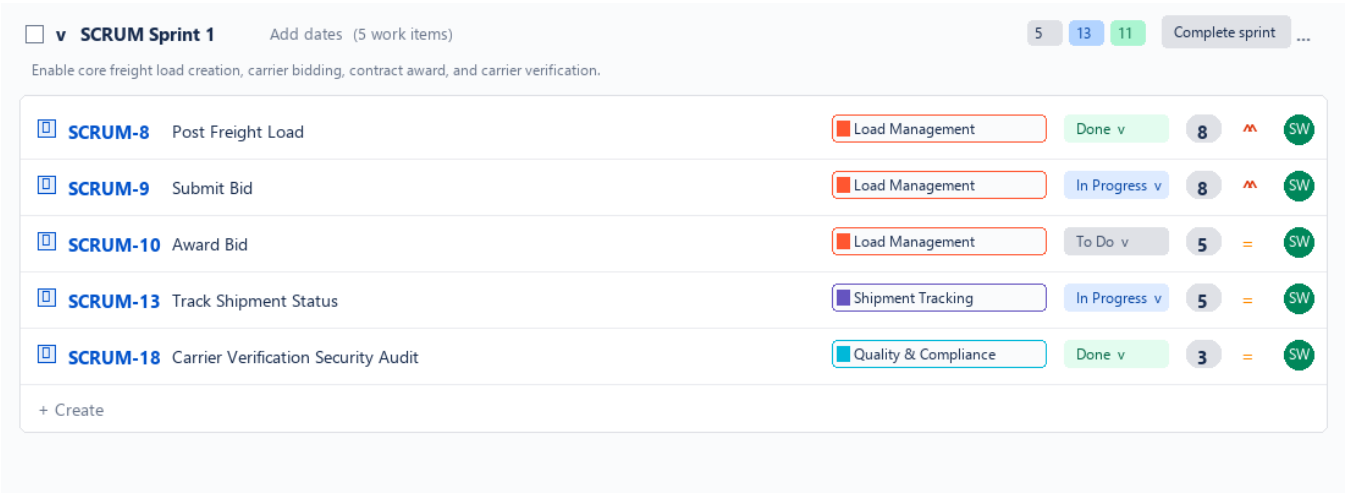


Figure 9: Sprint 1 active board showing in-flight progress across development tasks.

Sprint 1 — Completion

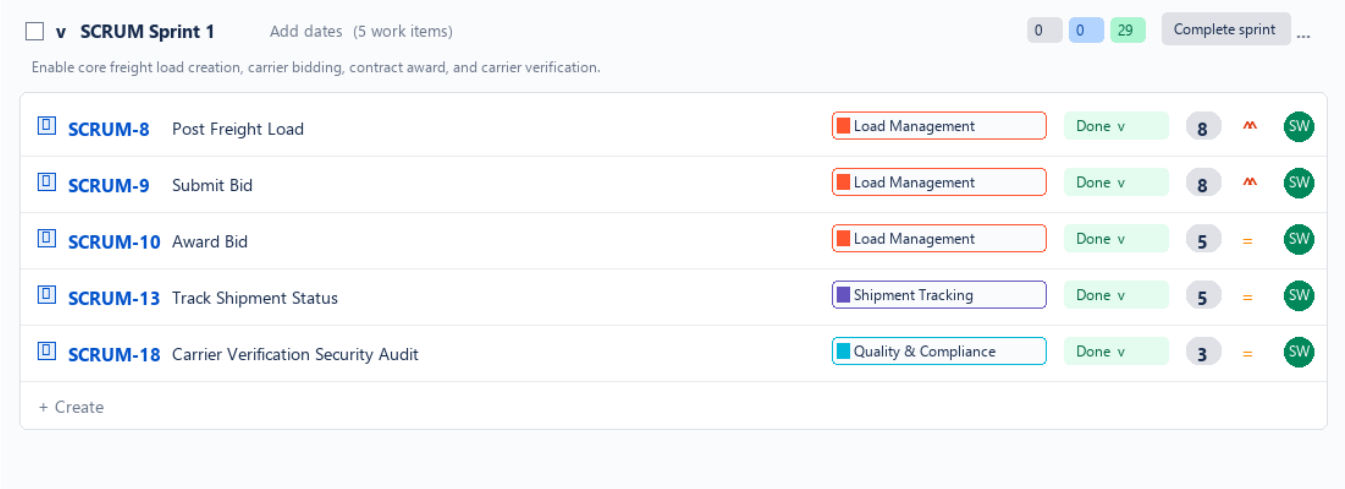


Figure 10: Sprint 1 completed with all 29 story points moved to Done.

Sprint 2 — Planning

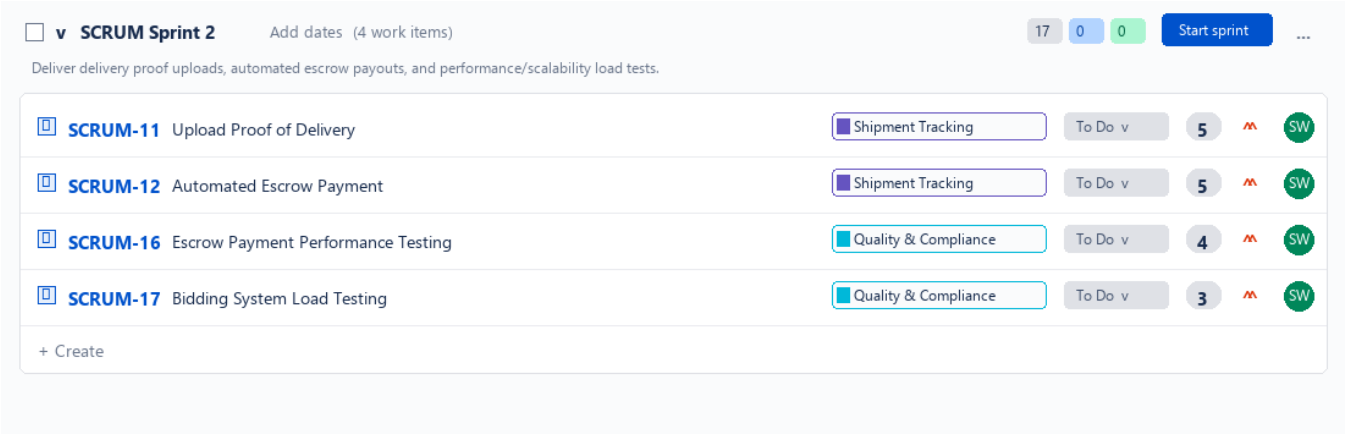


Figure 11: Sprint 2 scope: delivery verification, automated escrow payout, and scalability tests (17 story points).

Sprint 2 — To Do

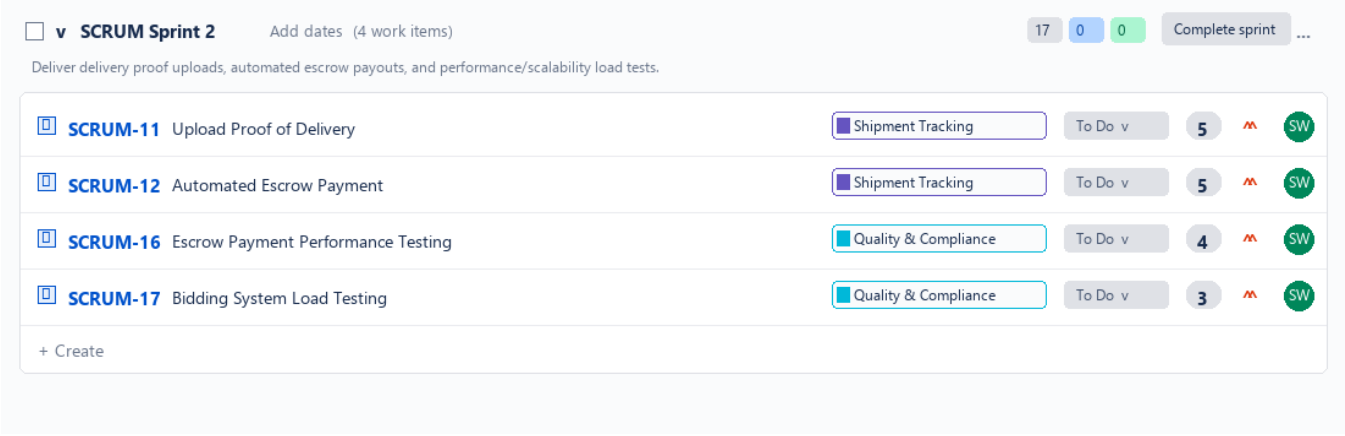


Figure 12: Sprint 2 active board at kickoff in To Do state.

Sprint 2 — Progression

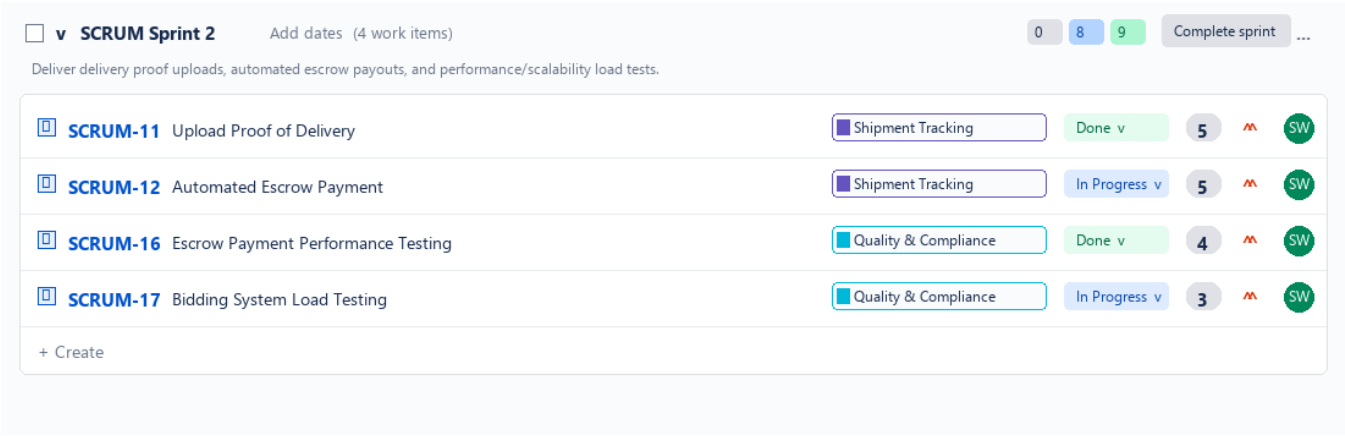


Figure 13: Sprint 2 in-progress execution and load benchmarking.

Sprint 2 — Completion

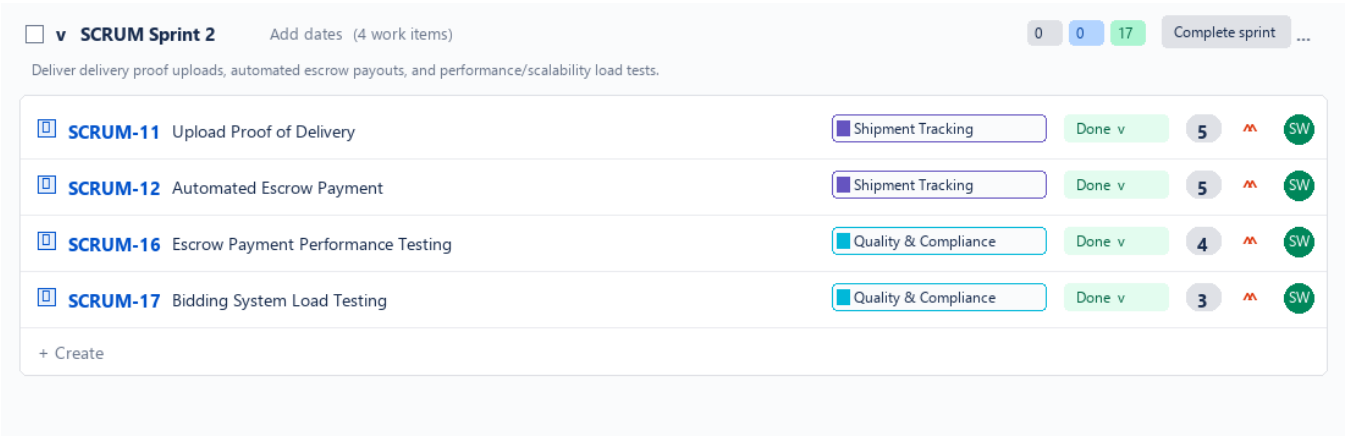


Figure 14: Sprint 2 completion: 100% of product backlog delivered (17 / 17 story points).

Sprint 1 Burndown Chart

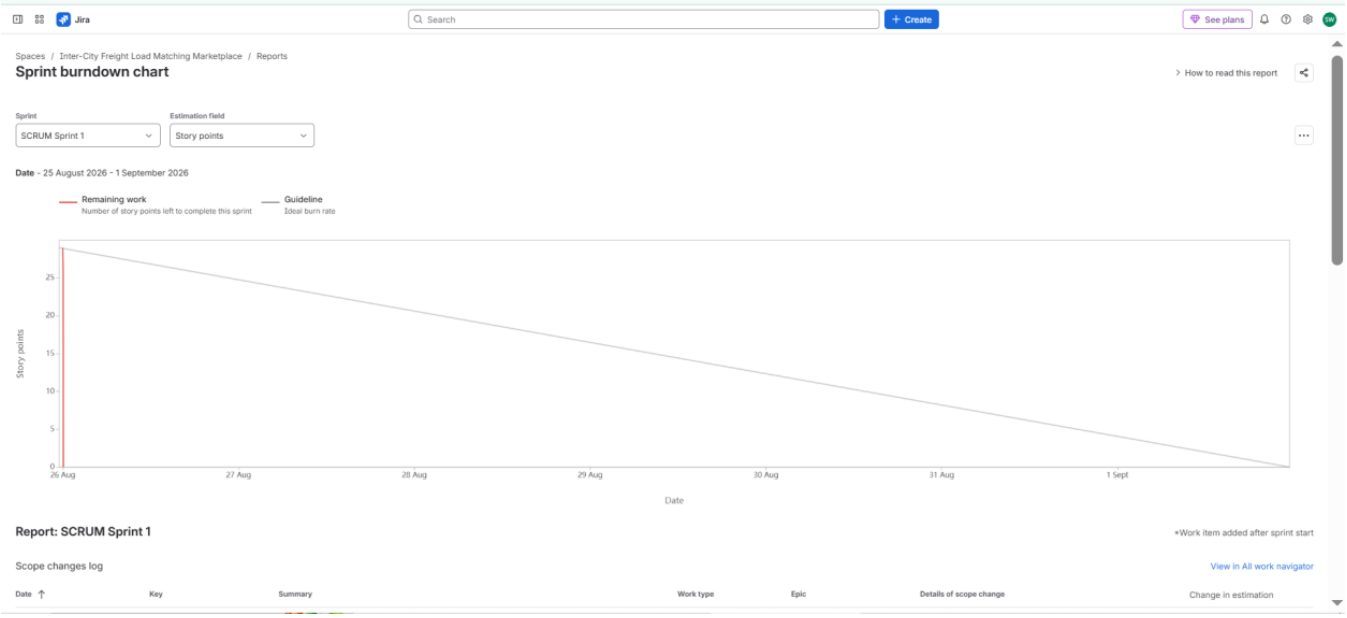


Figure 15: Jira Sprint 1 Burndown Chart (25 August - 1 September 2026, 29 SP).

Sprint 2 Burndown Chart

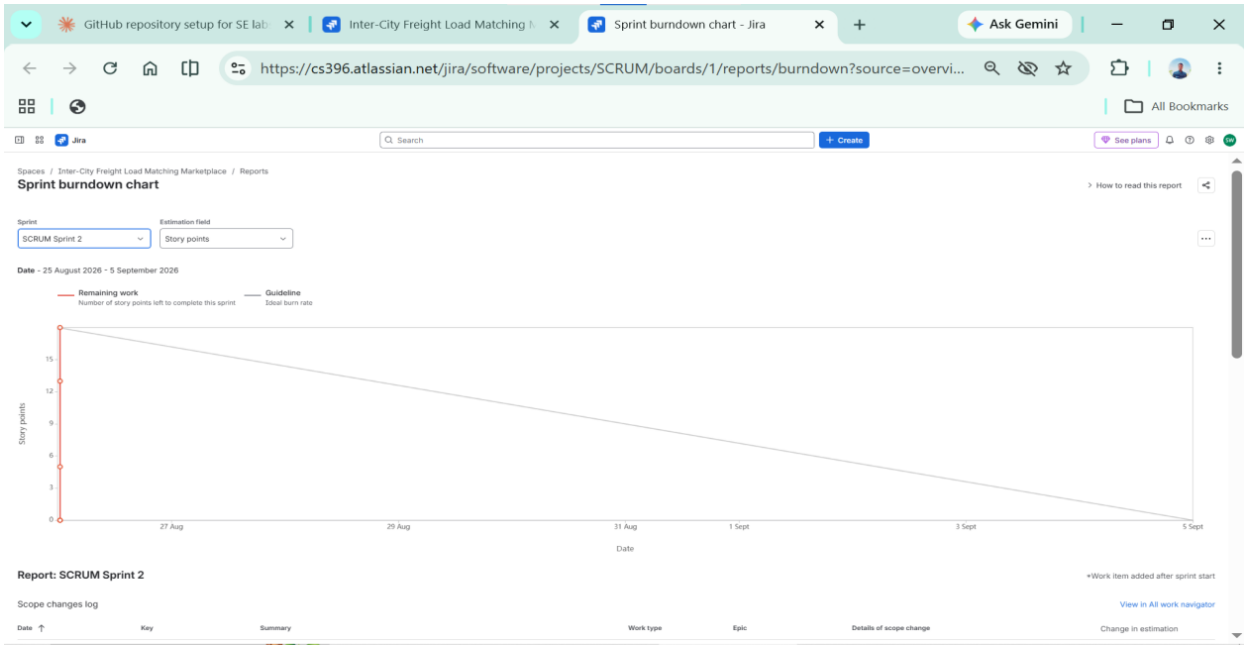


Figure 16: Jira Sprint 2 Burndown Chart (25 August - 5 September 2026, 17 SP).

Reflection Questions

1. Did your estimations reflect the actual effort?

The story point estimations accurately reflected the relative technical complexity and implementation risk of each feature. Larger user stories—such as Post Freight Load (SCRUM-8, 8 SP) and Submit Bid (SCRUM-9, 8 SP)—were appropriately sized with higher story points due to complex requirements including real-time auction synchronization, geographic route calculations, cargo volume/weight validations, and dynamic permit checks. Moderate complexity stories like Award Bid (SCRUM-10, 5 SP), Track Shipment Status (SCRUM-13, 5 SP), Upload Proof of Delivery (SCRUM-11, 5 SP), and Automated Escrow Payment (SCRUM-12, 5 SP) involved discrete state transitions, cryptographic signatures, and payment webhooks. Targeted verification and load testing tasks (SCRUM-16, SCRUM-17, SCRUM-18) were sized at 3–4 points as they utilized established benchmarking harnesses and external API contracts. In a simulated Scrum environment, Fibonacci estimation effectively calibrated team capacity and priority without conflating abstract complexity with raw developer hours.

2. Was your backlog well-prioritized?

Yes. The backlog prioritization strictly adhered to foundational functional dependencies and business risk. An inter-city freight marketplace cannot operate without cargo listings and carrier bidding; thus, Epic 1 was assigned High priority for Sprint 1. Real-time status tracking (SCRUM-13) and carrier permit verification (SCRUM-18) were prioritized alongside bidding to guarantee marketplace security and auction transparency from day one. Downstream fulfillment capabilities—such as proof-of-delivery upload (SCRUM-11) and automated escrow release (SCRUM-12)—along with automated load testing (SCRUM-16, SCRUM-17) were assigned to Sprint 2 because settlement and load testing logically depend upon an existing, awarded transport contract and functioning bidding engine developed in Sprint 1.

3. How did your simulated sprint align with your plan?

The simulated sprint aligned exceptionally well with the release plan. Sprint 1 absorbed 29 story points covering core marketplace mechanics, and all 5 planned stories transitioned systematically from To Do to In Progress and Done within the designated timebox. Sprint 2 absorbed the remaining 17 story points, concentrating on delivery verification, escrow payouts, and high-concurrency load testing, finishing with 100% backlog burndown and zero spillover debt. The sequential simulation accurately reflected Agile ceremonies (backlog grooming, sprint planning, and reviews) while validating that sprint commitments matched realistic team capacity.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts provided transparent operational insights into sprint delivery rhythm, team velocity, and workload progression. In Sprint 1, the chart demonstrated an ideal stepwise burn rate from 29 story points down to 0, validating that user stories were decoupled and progressed smoothly through review without blocking bottlenecks. In Sprint 2, the rapid closure of verification and testing tasks demonstrated high efficiency once core services were stable. For a Scrum team, steady downward burndown confirms dependable capacity planning, whereas early plateaus or sudden cliff drops highlight hidden dependencies, scope creep, or oversized stories requiring future backlog refinement.